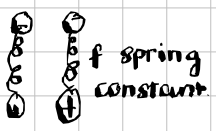
Oscillator Model

To understand material's response, it is useful to understand oscillator's model
Assume the net charge system is 0 and applied by an external field E



When there is a spring, we call Lorentz

$$m\ddot{x} = -fx - \frac{m}{\tau}\dot{x} - qE \Rightarrow \ddot{x} + \frac{1}{\tau}\dot{x} + \omega_0^2 x = -\frac{qE}{m} \qquad \omega_0^2 = f/m \quad 2\gamma = \frac{1}{\tau}$$



When the electron is free, we call Drude model $f = 0$

$$m\ddot{x} = -\frac{m}{\tau}\dot{x} - qE \Rightarrow \ddot{x} + \frac{1}{\tau}\dot{x} = -\frac{qE}{m}$$

The force by f is due to the binding interaction between electrons and protons. The friction γ is due to factors such as electrons scattering, interactions with holes, etc...
Using either FT, one arrives:

$$d(\omega) = -qx(\omega) = -\frac{q^2 E(\omega)}{m}\frac{1}{\omega^2 - \omega_0^2 + i\omega\frac{1}{\tau}}$$

With $P(\omega) = \chi(\omega)E(\omega)$, we have the response

$$\chi(\omega) = -\frac{Nq^2}{m}\frac{1}{\omega^2 - \omega_0^2 + i\omega\gamma} = -\frac{Nq^2}{m}\frac{1}{2\omega_0'}\left[\frac{1}{\omega - \omega_0' + i\gamma} - \frac{1}{\omega + \omega_0' + i\gamma}\right] \qquad \omega_0' = \sqrt{\omega_0^2 - \gamma^2}$$

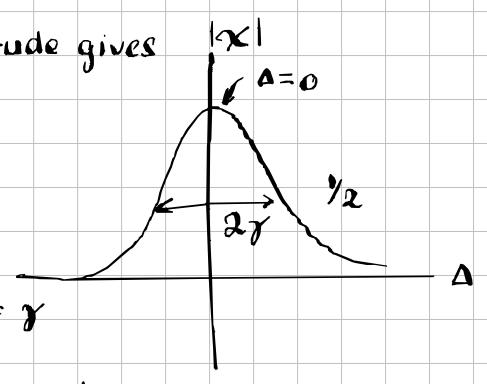
This form contains so much information.
Let $\Delta = \omega - \omega_0'$, we see immediately <u>poles</u>

$$\Delta + i\gamma = 0 \Rightarrow \omega = \omega_0' - i\gamma$$

The single number contains two pieces of physics

$$Re(\omega) = \omega_0' \quad Im(\omega) = -\gamma \Rightarrow e^{-i(\omega_0' - i\gamma)t} = e^{-i\omega_0' t}e^{-\gamma t}$$

an oscillation at ω_0' with a decay of γ
The poles are unachievable. Realistically, we scan "ω" as the real axis for response while γ deviates from the real axis
Now, look at $\chi(\omega) = |\chi(\omega)|e^{i\theta(\omega)}$, the amplitude gives



$$|\chi|^2 \propto \frac{1}{\Delta^2 + \gamma^2}$$ Lorentzian

At $\Delta = 0$, $|\chi|^2 \propto \frac{1}{\gamma^2}$ $\Rightarrow$ maximum

At half maximum, $\frac{1}{\Delta^2 + \gamma^2} = \frac{1}{2}\frac{1}{\gamma^2} \Rightarrow |\Delta| = \gamma$

The full width at half maximum $FWHM = 2\gamma = \frac{1}{\tau}$

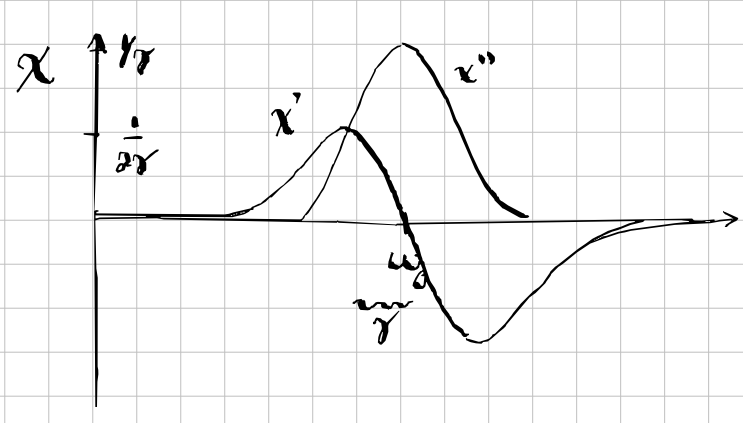
The phase is induced such that the created polarization is out of phase. Leading to a bad interference of induced E field and incoming E field.

$$\phi(\omega) = -\arctan\left(\frac{\gamma}{\Delta}\right) = \arctan\left(\frac{\chi''}{\chi'}\right)$$

At $\Delta = 0$, the shift $\phi(\omega) = -\pi/2$ $\Rightarrow$ exactly the cause of delay

The same form can be seen in a different form

$$\frac{1}{\Delta + i\gamma} = \frac{\chi'}{A}\frac{A}{\Delta^2 + \gamma^2} - \frac{\chi''}{i\gamma}\frac{i\gamma}{\Delta^2 + \gamma^2}$$



We call χ' as dispersion and
χ'' as absorption, but one needs care.
It is wrong to say χ' – phase only and χ'' amplitude.
Take $E(t) = E_0\cos(\omega t)$

$$P(\omega) = \chi(\omega)E(\omega)$$
$$= E_0[\chi'\cos\omega t + \chi''\sin\omega t]$$

The χ' is in phase with $E(t)$, χ'' is out of phase
χ' describes how materials absorp and relieve energy. In one cycle, no net gain.
But, it "rescales" $\varepsilon_0 \to \varepsilon_0 + \chi'$ amplitude.
The term χ'' removes energy from the field. $E\frac{dP}{dt}$

$$E\frac{dP}{dt} = E_0^2\omega\left[-\chi'\cos\cancel{\omega t}^{0}\sin\omega t + \chi''\cos^2\omega t\right]$$

These are real reasons not sth as $e^{i\chi'}e^{-\chi''}$

* When $\omega \approx \omega_0$, a large absorption occurs
* When χ' and $\chi'' \to 0$ $(\omega \to \infty)$, there is no physical response. The dipoles cannot follow the E field
* When $\omega << \omega_0$, this is normal dispersion. Electrons can follow the driven field really well. As ω is increasing, it is getting closer to ω_0 where material response is getting stronger

$$\chi' \uparrow \quad \varepsilon' \uparrow \quad \frac{d\varepsilon'}{d\omega}, \frac{dn}{d\omega} > 0$$

* When $\omega \approx \omega_0$, this region is anomalous dispersion $\frac{dn}{d\omega} < 0$

$$\chi' \downarrow \Rightarrow \varepsilon' \downarrow \quad \text{and} \quad \chi'' \uparrow\downarrow \Rightarrow \varepsilon'' \uparrow\downarrow$$

since $n^2 - \kappa^2 = \varepsilon' \qquad 2n\kappa = \varepsilon''$
light gets more absorbed, but refractive index is decreasing

* Just above resonance, $\chi' < 0$ such that there can be a case where

$$\varepsilon' = 1 + \chi' < 0 \quad n = 0 \quad \tilde{n} = i\kappa \quad (\text{when } \varepsilon'' = 0)$$

The field is simply decaying exponentially without propagating.

$$r = \frac{1 - \tilde{n}}{1 + \tilde{n}} \qquad R = |r|^2 = 1$$

An evanescent field in the material with perfect reflection

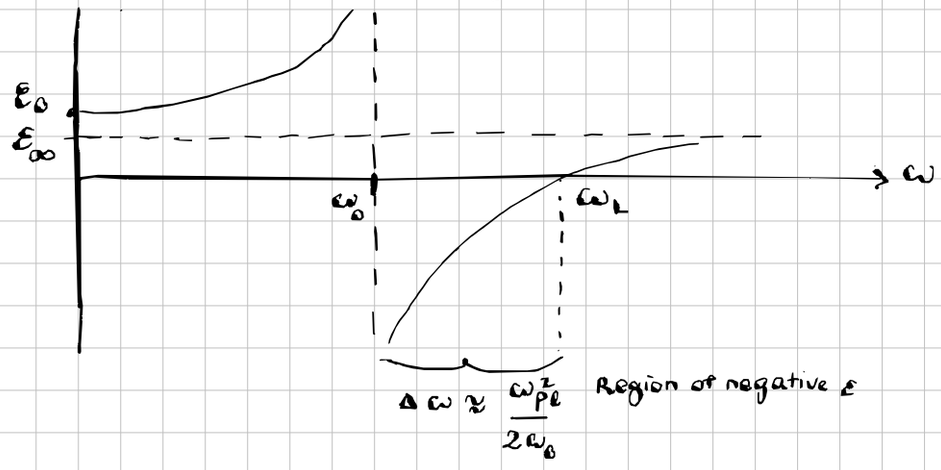
Zero Damping Limit

From $\chi(\omega)$, taking the limit $\gamma \to 0$ (no decay / scattering)

$$\chi(\omega) = \frac{Ne^2/m}{\omega_0^2 - \omega^2} \Rightarrow \varepsilon(\omega) = 1 + 4\pi \frac{Ne^2/m}{\omega_0^2 - \omega^2}$$

By defining $\omega_{pe}^2 = \frac{4\pi Ne^2}{m}$, we obtain

$$\varepsilon(\omega) = 1 + \frac{\omega_{pe}^2}{\omega_0^2 - \omega^2}$$



When $\omega < \omega_0$, $\omega_0^2 - \omega^2 > 0$, $\varepsilon > 1$

As $\omega \to \omega_0$, $\varepsilon \to \infty$

Immediately above ω_0, $\omega_0^2 - \omega^2 < 0$, $\varepsilon \to -\infty$

The permittivity is 0 when

$$\omega^2 = \omega_0^2 + \omega_{pe}^2 = \omega_L^2$$

The region between ω_0 and ω_L is negative If $\omega_{pe} << \omega_0$

$$\omega_L = \sqrt{\omega_0^2 + \omega_{pe}^2} \approx \omega_0 + \frac{\omega_{pe}^2}{2\omega_0}$$

$$\Delta\omega = \frac{\omega_{pe}^2}{2\omega_0}$$

Instead of assuming high frequency background as one

$$\varepsilon(\omega) = \varepsilon_\infty + \frac{S}{\omega_T^2 - \omega^2} \qquad \varepsilon(0) = \varepsilon_\infty + \frac{S}{\omega_T^2} = \varepsilon_0 \quad S = (\varepsilon_0 - \varepsilon_\infty)\omega_T^2$$

$$\varepsilon(\omega) = \varepsilon_\infty + \frac{(\varepsilon_0 - \varepsilon_\infty)}{1 - \omega^2/\omega_T^2}$$

The transverse frequency

$$\varepsilon(\omega_T) = \pm\infty \qquad \text{in our case } \omega_T = \omega_0$$

Consider

$$0 = \varepsilon(\omega_L) = \varepsilon_\infty + \frac{(\varepsilon_0 - \varepsilon_\infty)}{1 - \omega^2/\omega_T^2}$$

$$\boxed{\frac{\varepsilon_0}{\varepsilon_\infty} = \frac{\omega_L^2}{\omega_T^2}}$$ Lyddane - Sachs - Teller Relation

Normally $\varepsilon_0 > \varepsilon_\infty$, $\omega_L > \omega_T$. The regime $\omega_T < \omega < \omega_L$: $\varepsilon < 0$ allows $\tilde{n}^2 = \varepsilon < 0$, $\tilde{n} = i\kappa$ allow strong reflection

Optical Response of a Metal.

Each type of electron adds $P_i = \chi_i E$ Therefore $P = (\sum_i \chi_i) E$

The permittivity becomes

$$\varepsilon = 1 + 4\pi \sum_i \chi_i$$

For the core / bound electrons, they have a restoring force $\omega_{0,core} \neq 0$

The conduction electrons are free $\omega_{0,core} = 0$

By ignoring damping, we arrive at

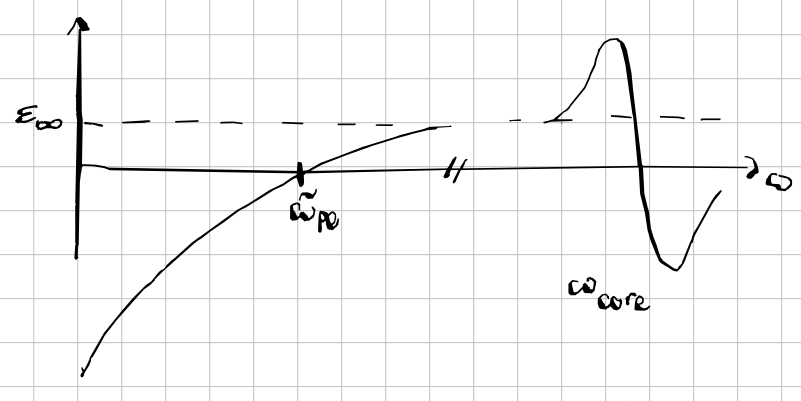
$$\varepsilon(\omega) = 1 - \frac{\omega_{p,core}^2}{\omega^2 - \omega_{0,core}^2} - \frac{\omega_{p,cond}^2}{\omega^2}$$

Suppose we are well below the core electron resonance.

$$\omega \ll \omega_{0,core} \quad - \frac{\omega_{p,core}^2}{\omega^2 - \omega_{0,core}^2} \approx \frac{\omega_{p,core}^2}{\omega_{0,core}^2}$$

Absorb into a background dielectric constant.

$$\varepsilon_\infty \equiv 1 + \frac{\omega_{p,core}^2}{\omega_{0,core}^2} \qquad \varepsilon(\omega) = \varepsilon_\infty - \frac{\omega_p^2}{\omega^2} = \varepsilon_\infty \left(1 - \frac{\tilde{\omega}_p^2}{\omega^2}\right) \qquad \tilde{\omega}_p = \frac{\omega_p}{\sqrt{\varepsilon_\infty}}$$



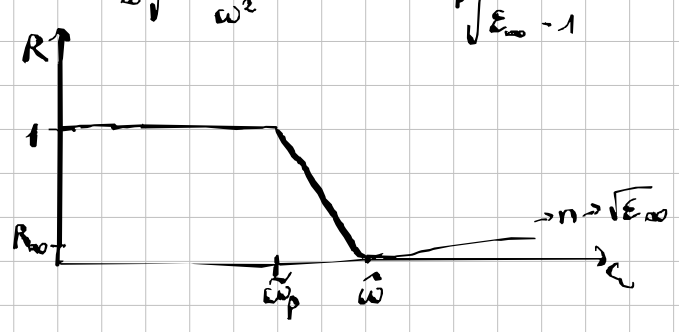
At $\omega = \tilde{\omega}_p$, $\varepsilon = 0$, for $\omega < \tilde{\omega}_p$, $\varepsilon < 0$ $\quad \tilde{n}^2 = \varepsilon \Rightarrow \tilde{n} = ik$

The field is then evanescent. Now

$$R = \left(\frac{n-1}{n+1}\right)^2 \qquad R = \left|\frac{\tilde{n}-1}{\tilde{n}+1}\right|^2 \Rightarrow n = \sqrt{\varepsilon_\infty}\sqrt{1 - \frac{\tilde{\omega}_p^2}{\omega^2}}$$

Reflectivity is 0 when $n = 1$

$$1 = \sqrt{\varepsilon_\infty}\sqrt{1 - \frac{\tilde{\omega}_p^2}{\omega^2}} \Rightarrow \hat{\omega} = \tilde{\omega}_p \sqrt{\frac{\varepsilon_\infty}{\varepsilon_\infty - 1}}$$



If $\tilde{\omega}_p > \omega_{visible}$ → metal reflects visible light

If $\omega >> \tilde{\omega}_{pe}$, metal becomes transparent (UV transparency)

Now, we add collisions

$$m\ddot{x} + m\gamma\dot{x} = -eE$$

The Drude dialectric becomes

$$\varepsilon(\omega) = \varepsilon_\infty - \frac{\omega_p^2}{\omega(\omega + i\gamma)}$$

The metal also absorbs. Electron-phonon scattering and impurities are major sources of momentum relaxation. ordinary electron electron collisions conserve total momentum and therefore do not by themselves give DC.

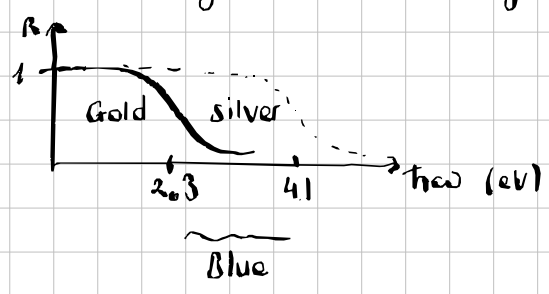
Drude model tells why metals generate low-frequency light

free electrons → $\varepsilon' < 0$ → strong reflections.

But real metals filled electronic bands. Interband transition for gold

$$\hbar\omega \sim 2.3 eV$$

These transitions strongly absorb blue light. Gold reflects red and yellow while absorbing blue.



For free electrons, $\omega_p = \sqrt{\frac{ne^2}{\varepsilon_0 m}}$ n is carrier density

Define free space plasma wavelength

$$\lambda_p = \frac{2\pi c}{\omega_p} \qquad \omega_p \propto \sqrt{n} \qquad \lambda_p \propto 1/\sqrt{n}$$

For a typical metal, $n \sim 10^{22}\ cm^{-3}$

$$\omega_p \sim 5.7 e15\ rad/s \qquad \lambda_p = \frac{2\pi c}{\omega_p} = .33 \mu m$$

For a doped semiconductor, $n \sim 10^{8}\ cm^{-3}$

$$\omega_p \sim 5.7 e13\ rad/s \qquad \lambda_p = 30 \mu m$$

More free electrons mean the electron gas can collectively oscillate much faster

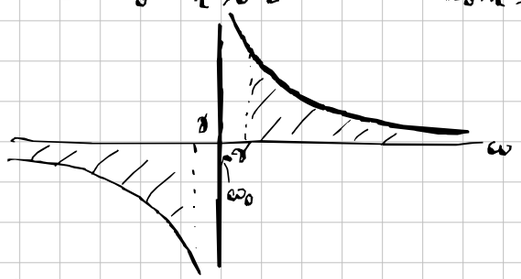
Interdependence of Absorption and Refractive Index

Absorption and refractive index dispersion are not independent
Consider Sokhotski - Plemelj density

$$\lim_{\gamma\to 0^+} \frac{1}{x+i\gamma} = \mathcal{P}\frac{1}{x} - i\pi\delta(x) \Rightarrow \frac{1}{\omega-\omega_0+i0^+} = \mathcal{P}\frac{1}{\omega-\omega_0} - i\pi\delta(\omega-\omega_0)$$

The first term is dispersive real part and second term is absorptive imaginary part

$$\mathcal{P}\int d\omega \frac{G(\omega)}{\omega-\omega_0} = \lim_{\eta\to 0}\left[\int^{\omega_0-\eta} + \int_{\omega_0+\eta}\right]\frac{G(\omega)}{\omega-\omega_0}\,d\omega \qquad \mathcal{P} \text{ excludes divergence point}$$



Kramers Kronig Relation

$$\chi(\omega) = \chi'(\omega) + i\chi''(\omega)$$

The relation reads

$$\chi'(\omega) = \frac{1}{\pi}\mathcal{P}\int_{-\infty}^{\infty}\frac{\chi''(\Omega)}{\Omega-\omega}\,d\Omega \qquad \chi''(\omega) = -\frac{1}{\pi}\mathcal{P}\int_{-\infty}^{\infty}\frac{\chi'(\Omega)}{\Omega-\omega}\,d\Omega$$

So knowing all frequencies determine the other
This is causality: material cannot respond before the electric field is applied.
In zero damping limit, the resonance contribution has poles at $\pm\omega_0$
The identity turns each pole into

$$\chi(\omega) = \frac{Ne^2}{m}\left(\frac{1}{\omega-\omega_0+i\gamma} - \frac{1}{\omega+\omega_0+i\gamma}\right)$$

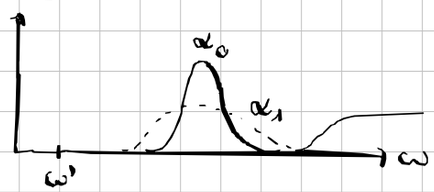
$$\xrightarrow[\gamma\to 0]{} \frac{Ne^2}{m}\left(\mathcal{P}\frac{1}{\omega-\omega_0} - \mathcal{P}\frac{1}{\omega+\omega_0}\right) - i\pi\left[\delta(\omega-\omega_0)-\delta(\omega+\omega_0)\right]$$

At a pole $\qquad \chi' \sim \mathcal{P}\frac{1}{\omega-\omega_0} \qquad \chi'' \sim \delta(\omega-\omega_0)$

A very sharp absorption line produces a dispersive feature extending to frequencies on both sides of the line.

Suppose something modifies an absorption spectrum.

$$\alpha_0(\omega) \to \alpha_1(\omega) \qquad \Delta\alpha(\omega) = \alpha_1(\omega) - \alpha_0(\omega) \qquad \Delta n(\omega)?$$



$$\Delta\alpha(\omega) \simeq \frac{\omega}{cn_0}\Delta\varepsilon''(\omega) \qquad \Delta\varepsilon''(\omega) \simeq \frac{cn_0}{\omega}\Delta\alpha(\omega)$$

When Kramers Kronig applies on the change

$$\Delta\varepsilon'(\omega) = \frac{2}{\pi}\mathcal{P}\int_0^{\infty}\frac{\Omega\,\Delta\varepsilon''(\Omega)}{\Omega^2-\omega^2}\,d\Omega = \frac{2cn_0}{\pi}\mathcal{P}\int_0^{\infty}\frac{\Delta\alpha(\Omega)}{\Omega^2-\omega^2}\,d\Omega$$

Outside the absorption resonance, where loss is weak

$$n = \sqrt{\varepsilon}$$

For a small change, $\quad n = \sqrt{\varepsilon_0+\Delta\varepsilon'} \simeq n_0 + \frac{\Delta\varepsilon'}{2n_0} \qquad \Delta n \simeq \frac{\Delta\varepsilon'}{2n_0} \qquad \Delta n(\omega) \simeq \frac{c}{\pi}\mathcal{P}\int_0^{\infty}\frac{\Delta\alpha(\Omega)}{\Omega^2-\omega^2}\,d\Omega$

When changing absorption around resonance, it changes index away from resonance.